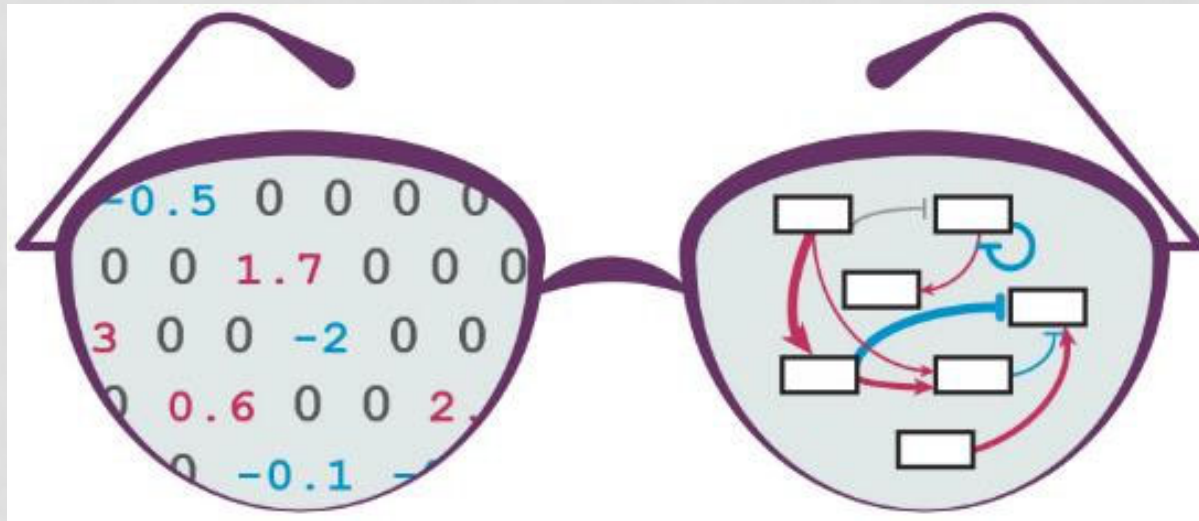




GRNSIGHT

NICOLE ANGUIANO

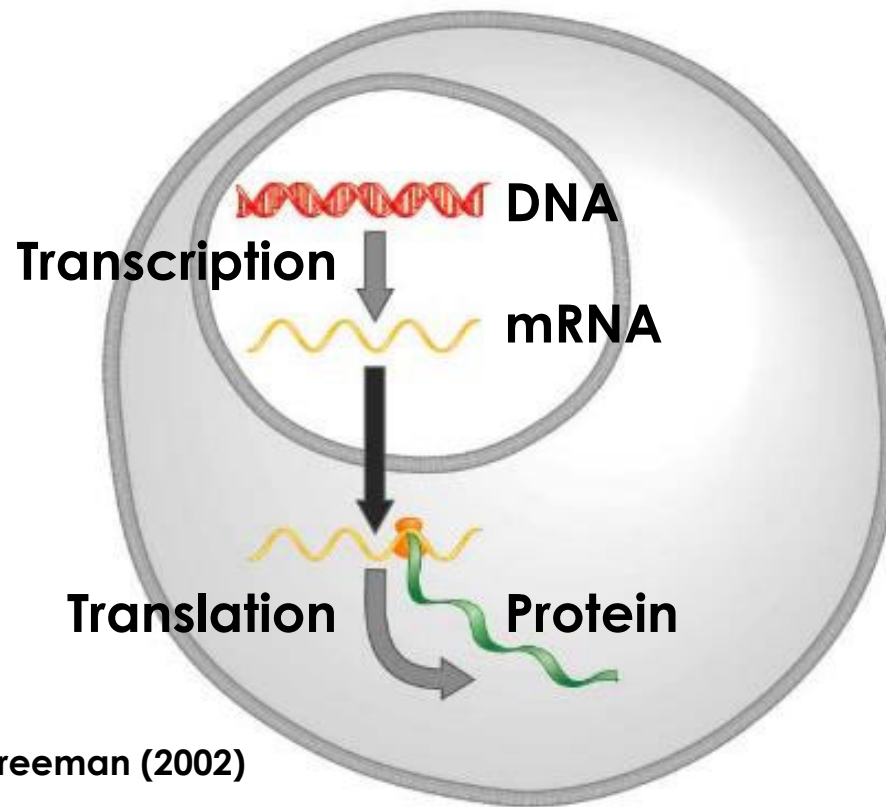
May 5,
2017

CMSI
402

OUTLINE

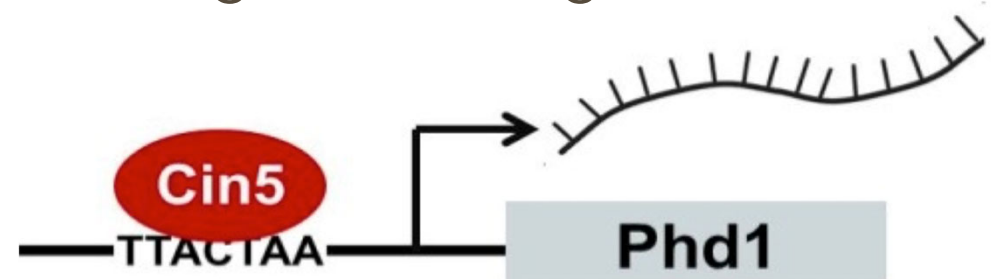
- GRNsight is a web app and service for visualizing small- to medium-scale gene regulatory networks (GRNs).
- The GRNsight project began in 2014 as a web app intended for viewing GRNs generated from the GRNmap mathematical modeling software.
- The primary goal for this project was to improve the graph layout by separating the visible area of the graph from the drawable area of the graph.
- The goals and resulting subgoals were achieved in version 2 of GRNsight.

A GENE REGULATORY NETWORK CONTROLS THE LEVEL OF EXPRESSION OF GENES AND PROTEINS

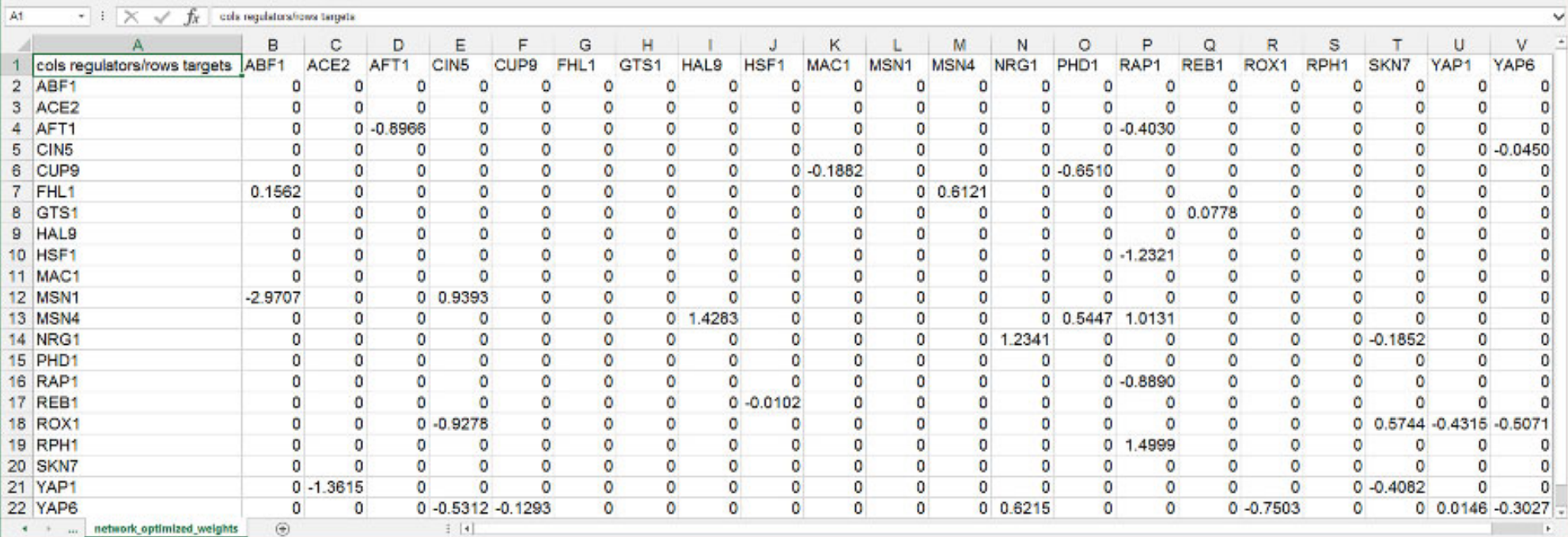


Freeman (2002)

- Mechanism of control: transcription factors
- Transcription factors bind to regulatory DNA sequences
 - Activators increase gene expression
 - Repressors decrease gene expression
- GRNmap model produces weight values representing the strength of the regulation



REPRESENTATIONS OF GRNS CAN BE DIFFICULT TO UNDERSTAND WITHOUT VISUALIZATION

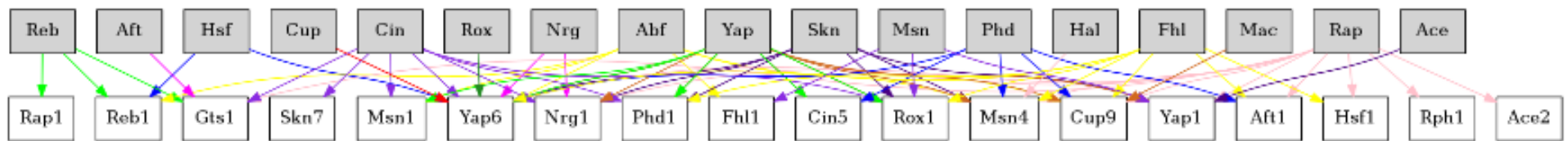


	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
1	cols regulators/rows targets	ABF1	ACE2	AFT1	CIN5	CUP9	FHL1	GTS1	HAL9	HSF1	MAC1	MSN1	MSN4	NRG1	PHD1	RAP1	REB1	ROX1	RPH1	SKN7	YAP1	YAP6
2	ABF1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
3	ACE2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	AFT1	0	0	-0.8966	0	0	0	0	0	0	0	0	0	0	0	-0.4030	0	0	0	0	0	0
5	CIN5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-0.0450
6	CUP9	0	0	0	0	0	0	0	0	0	-0.1882	0	0	0	-0.6510	0	0	0	0	0	0	0
7	FHL1	0.1562	0	0	0	0	0	0	0	0	0	0	0.6121	0	0	0	0	0	0	0	0	0
8	GTS1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0778	0	0	0	0	0
9	HAL9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
10	HSF1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1.2321	0	0	0	0	0	0
11	MAC1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
12	MSN1	-2.9707	0	0	0.9393	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
13	MSN4	0	0	0	0	0	0	0	1.4283	0	0	0	0	0	0.5447	1.0131	0	0	0	0	0	0
14	NRG1	0	0	0	0	0	0	0	0	0	0	0	0	1.2341	0	0	0	0	0	-0.1852	0	0
15	PHD1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
16	RAP1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-0.8890	0	0	0	0	0	0
17	REB1	0	0	0	0	0	0	0	0	-0.0102	0	0	0	0	0	0	0	0	0	0	0	0
18	ROX1	0	0	0	-0.9278	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.5744	-0.4315	-0.5071
19	RPH1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1.4999	0	0	0	0	0	0
20	SKN7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
21	YAP1	0	-1.3615	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-0.4082	0	0
22	YAP6	0	0	0	-0.5312	-0.1293	0	0	0	0	0	0	0	0.6215	0	0	0	-0.7503	0	0	0.0146	-0.3027

Weighted adjacency matrix in .xlsx format

EXISTING GRN VISUALIZATION SOFTWARE DID NOT FIT THE USE CASE NEEDED BY THE GRNMAP TEAM

- YEASTRACT GRN visualizations are static, difficult to interpret, and don't display edge weight values



- Cytoscape and Gephi must be installed, are optimized for large GRNs, and have complex features with a steep learning curve
- Making a graph by hand in Illustrator is time-consuming
- GRNsight exists as a web app, is optimized for small- to medium-scale graphs, is quick to customize, and displays weight values in an intuitive way for the user

GRNSIGHT TURNS THIS...

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
1	cols regulators/rows targets	ABF1	ACE2	AFT1	CIN5	CUP9	FHL1	GTS1	HAL9	HSF1	MAC1	MSN1	MSN4	NRG1	PHD1	RAP1	REB1	ROX1	RPH1	SKN7	YAP1	YAP6
2	ABF1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
3	ACE2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	AFT1	0	0	-0.8966	0	0	0	0	0	0	0	0	0	0	0	-0.4030	0	0	0	0	0	0
5	CIN5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-0.0450
6	CUP9	0	0	0	0	0	0	0	0	0	-0.1882	0	0	0	-0.6510	0	0	0	0	0	0	0
7	FHL1	0.1562	0	0	0	0	0	0	0	0	0	0	0.6121	0	0	0	0	0	0	0	0	0
8	GTS1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0778	0	0	0	0	0
9	HAL9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
10	HSF1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1.2321	0	0	0	0	0	0
11	MAC1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
12	MSN1	-2.9707	0	0	0.9393	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
13	MSN4	0	0	0	0	0	0	0	1.4283	0	0	0	0	0	0.5447	1.0131	0	0	0	0	0	0
14	NRG1	0	0	0	0	0	0	0	0	0	0	0	0	1.2341	0	0	0	0	0	-0.1852	0	0
15	PHD1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
16	RAP1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-0.8890	0	0	0	0	0	0
17	REB1	0	0	0	0	0	0	0	0	-0.0102	0	0	0	0	0	0	0	0	0	0	0	0
18	ROX1	0	0	0	-0.9278	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.5744	-0.4315	-0.5071
19	RPH1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1.4999	0	0	0	0	0	0
20	SKN7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
21	YAP1	0	-1.3615	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-0.4082	0	0
22	YAP6	0	0	0	-0.5312	-0.1293	0	0	0	0	0	0	0	0.6215	0	0	0	-0.7503	0	0	0.0146	-0.3027

Weighted adjacency matrix in .xlsx format

GRNSIGHT TURNS THIS...

	A	B	C
1	CIN5	0.9393	MSN1
2	CIN5	-0.9278	ROX1
3	CIN5	-0.5312	YAP6
4	CUP9	-0.1293	YAP6
5	HSF1	-0.0102	REB1
6	MSN4	0.6121	FHL1
7	NRG1	1.2341	NRG1
8	NRG1	0.6215	YAP6
9	RAP1	-1.2321	HSF1
10	RAP1	1.0131	MSN4
11	RAP1	-0.8890	RAP1
12	RAP1	-0.4030	AFT1
13	RAP1	1.4999	RPH1
14	AFT1	-0.8966	AFT1
15	REB1	0.0778	GTS1
16	ROX1	-0.7503	YAP6
17	YAP1	-0.4315	ROX1
18	YAP1	0.0146	YAP6
19	YAP6	-0.0450	CIN5
20	YAP6	-0.5071	ROX1
21	YAP6	-0.3027	YAP6
22	ABF1	0.1562	FHL1
23	ABF1	-2.9707	MSN1
24	ACE2	-1.3615	YAP1
25	HAL9	1.4283	MSN4
26	MAC1	-0.1882	CUP9
27	PHD1	-0.6510	CUP9
28	PHD1	0.5447	MSN4

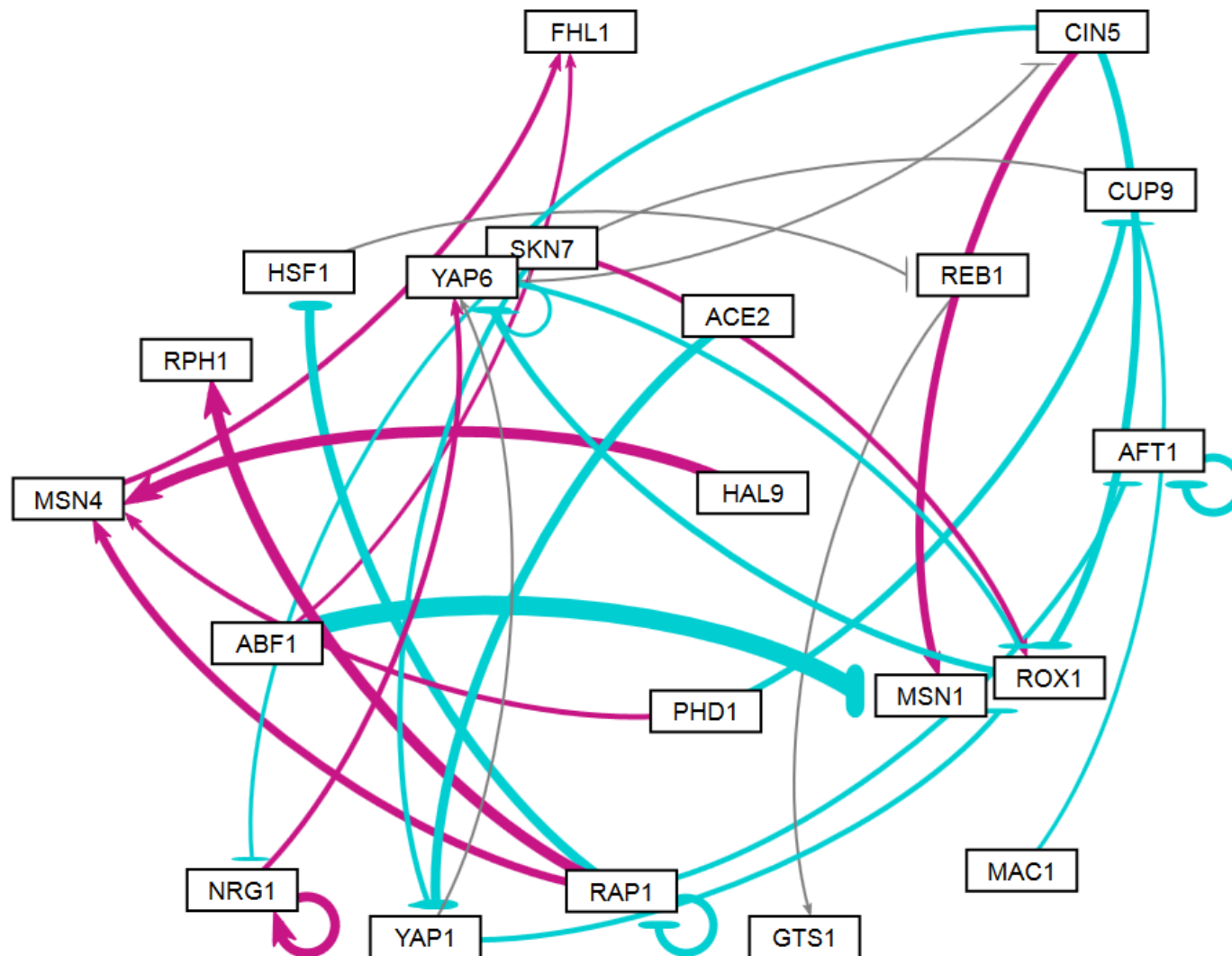
Weighted Simple Interaction Format (.sif) file

GRNSIGHT TURNS THIS...

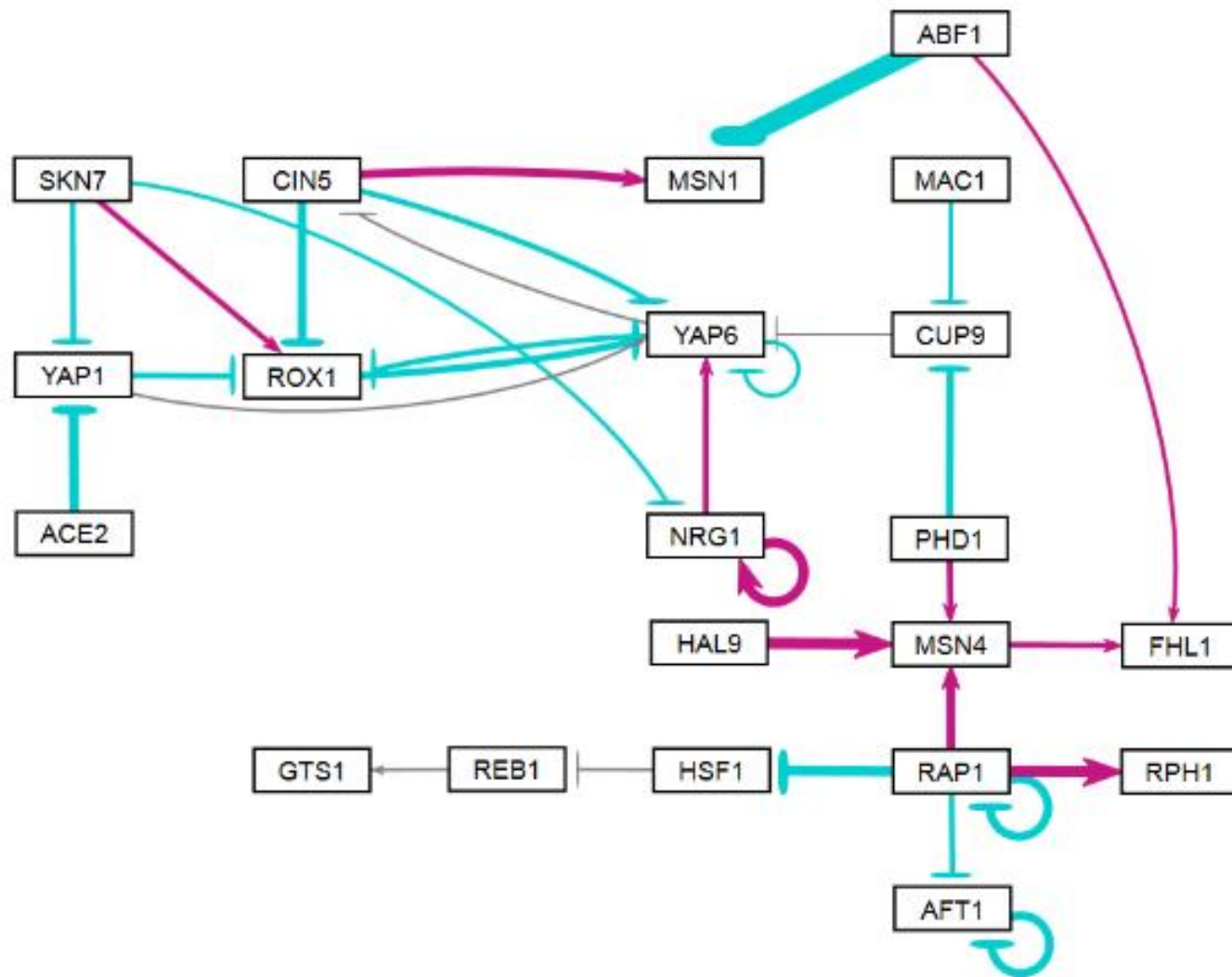
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  <!-- Exported by GRNsight v1.18 http://dondi.github.io/GRNsight/ https://github.com/dondi/GRNsight/releases/tag/v1.18 -->
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  <key id="interaction" for="edge" attr.name="interaction" attr.type="string"/>
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  <key id="weight" for="edge" attr.name="weight" attr.type="double"/>
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    <node id="ABF1">
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    <node id="ACE2">
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    </node>
    <node id="AFT1">
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      <data key="name">YAP1 (pd) YAP6</data>
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    </edge>
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      <data key="name">YAP6 (pd) YAP6</data>
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    </edge>
  </graph>
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```

Weighted GraphML file

INTO THIS



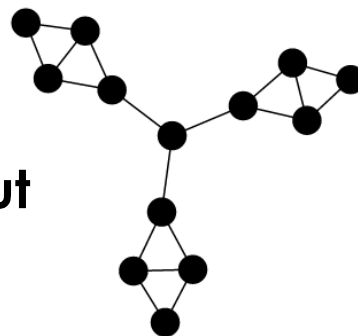
WHICH CAN BE CUSTOMIZED TO THIS!



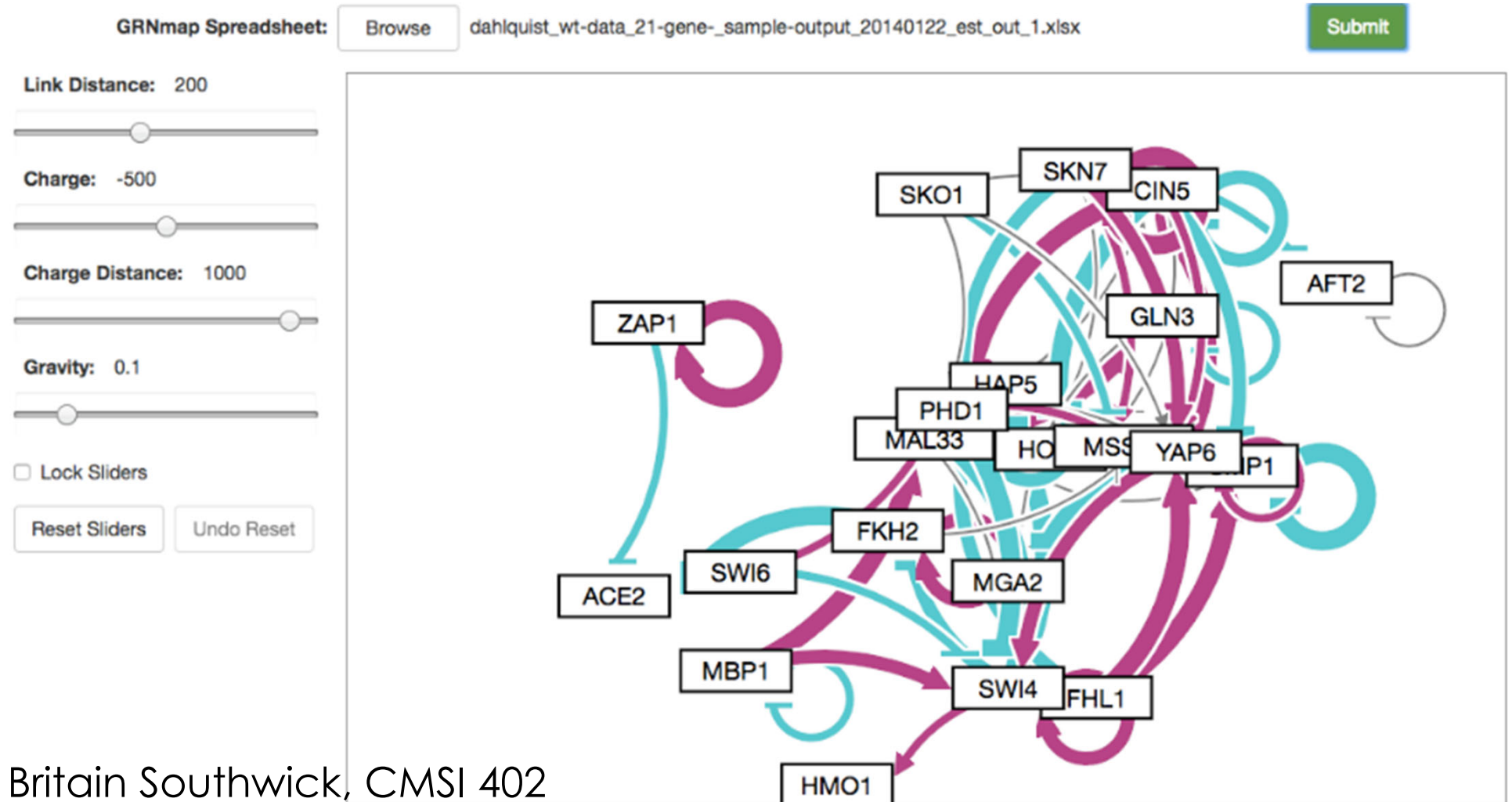
INITIAL GRNSIGHT IMPLEMENTATION REQUIRED CUSTOMIZATION OF THE D3 LIBRARY

- Developed as part of a 402 project by Britain Southwick
- Uses D3.js to generate a graph derived from input network data
- D3 dynamically manipulates HTML, CSS and SVG to form the elements of the graph
 - GRNsight implements D3's force layout, which applies a physics-based simulation to the graph

Default D3 layout

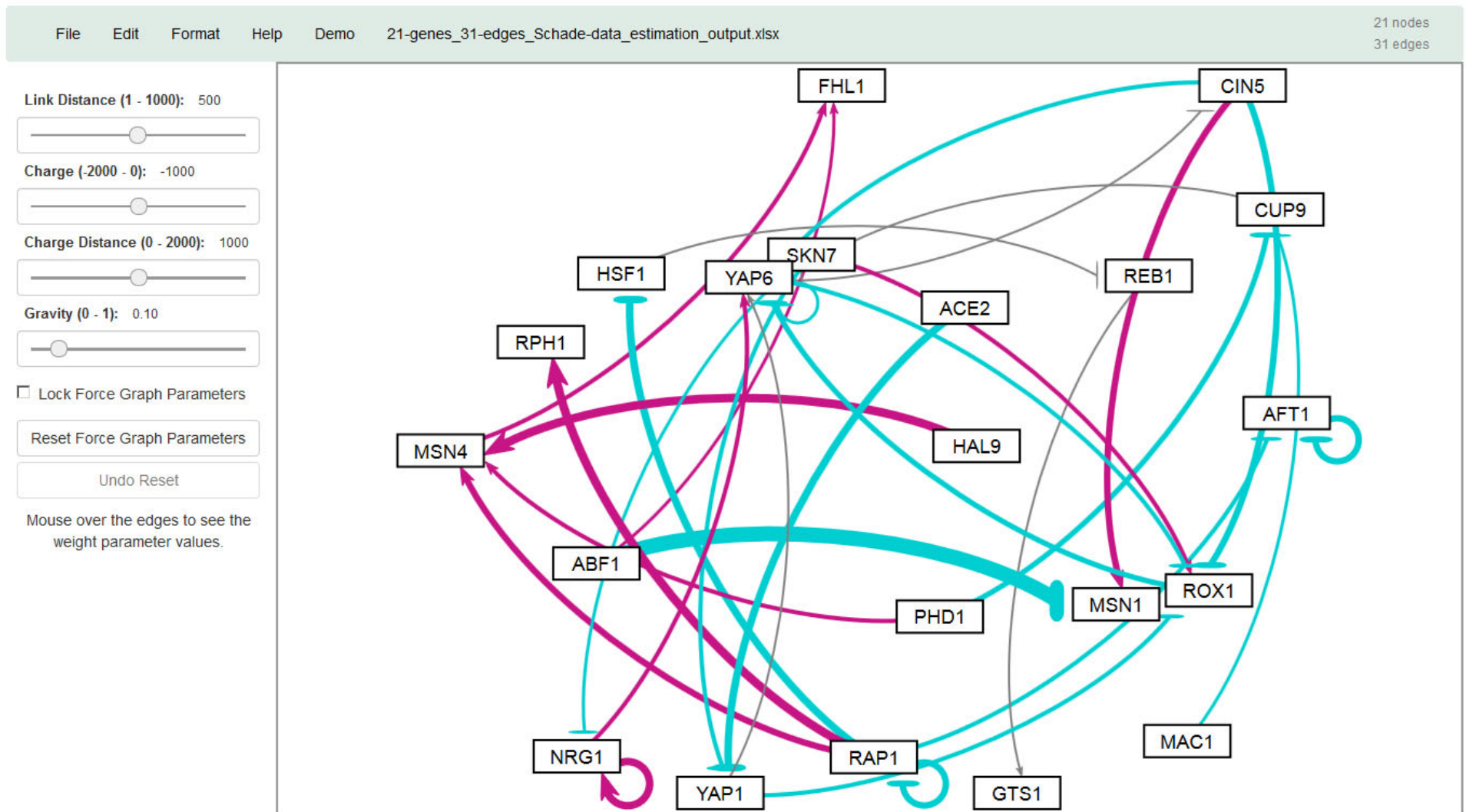


GRNSIGHT V0 (MAY 2014)



Britain Southwick, CMSI 402

GRNSIGHT V1 (AUGUST 2016)



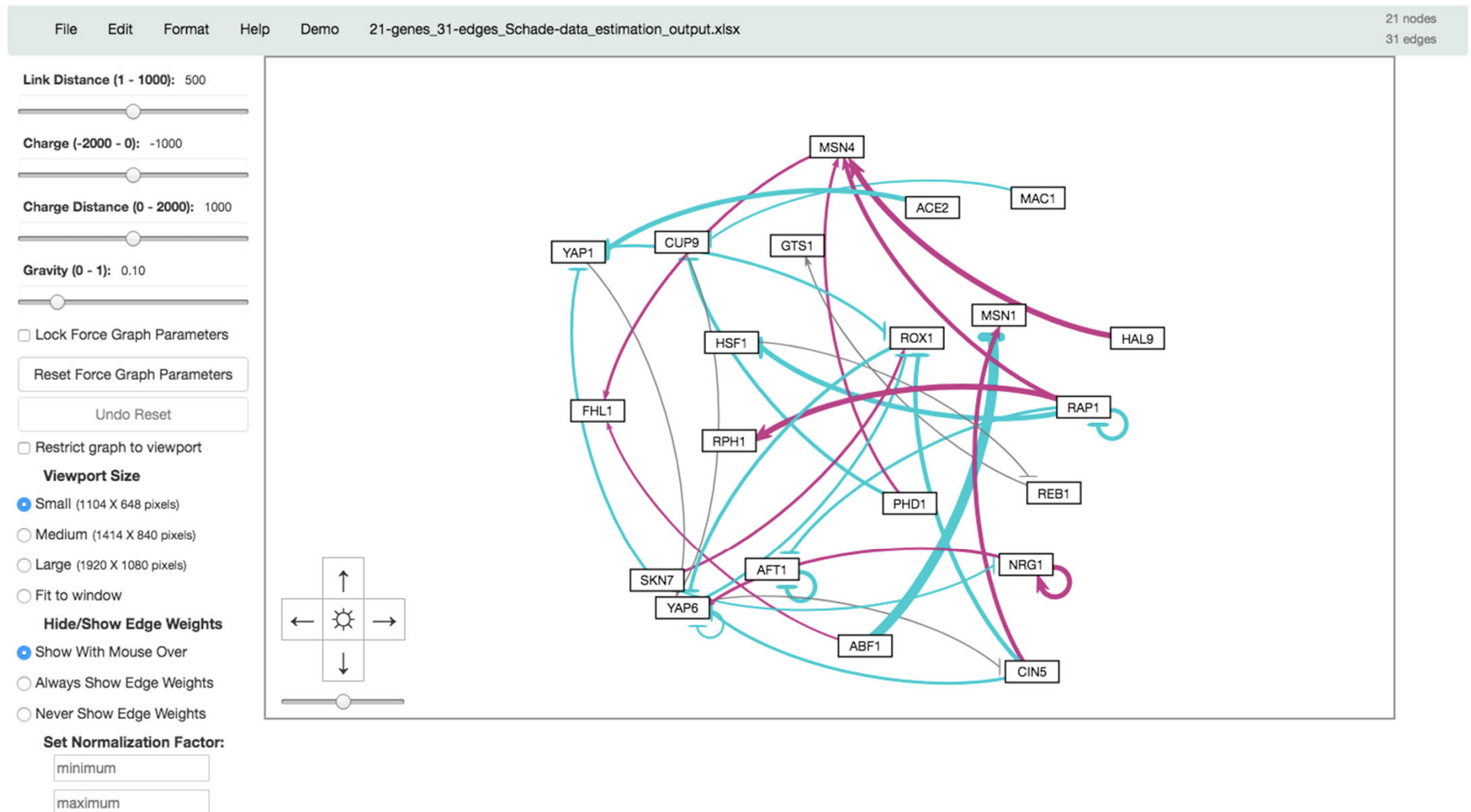
WITH THE INITIAL GRNSIGHT APPLICATION COMPLETE, IT WAS TIME FOR IMPROVEMENTS

- **Flaw:** Force graph could not fully relax when loaded
 - Viewport and bounding box were the same, and came in only one size
- **Solution:** Allow bounding box to expand as the force was applied
 - **Consequence:** Need a way to navigate the newly expanded bounding box
- **Goal:** Separate viewport from bounding box, and allow multiple sizes of both for different screens
 - Add zooming and scrolling navigation functions
- Smaller goals included quality of life improvements for the developers as well as code cleanup

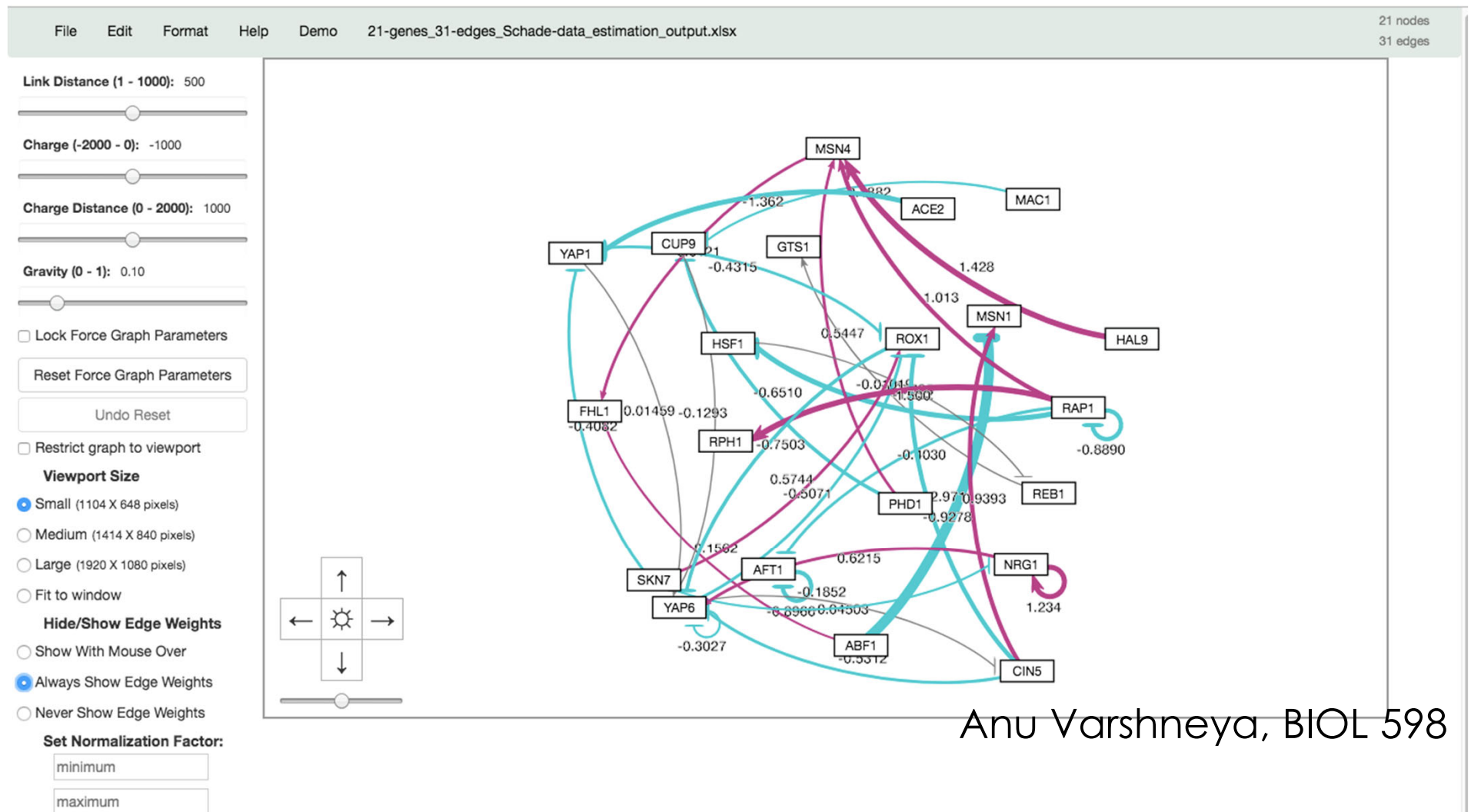
THE DECISION TO WORK ON GRNSIGHT FOR 402 WAS INSPIRED BY A NUMBER OF DIFFERENT FACTORS

- Allows me to contribute to an ongoing research project
 - Dahlquist Lab: Wet Lab, GRNmap, Data Analysis, GRNsight
- Combines my interest in biology with my skills as a computer scientist
- Create a maintainable project that the remaining members of the GRNsight team could continue without feeling too confused
- GRNsight was the first major coding project I ever did at LMU, and it felt fitting for it to also be the last one I did here as well

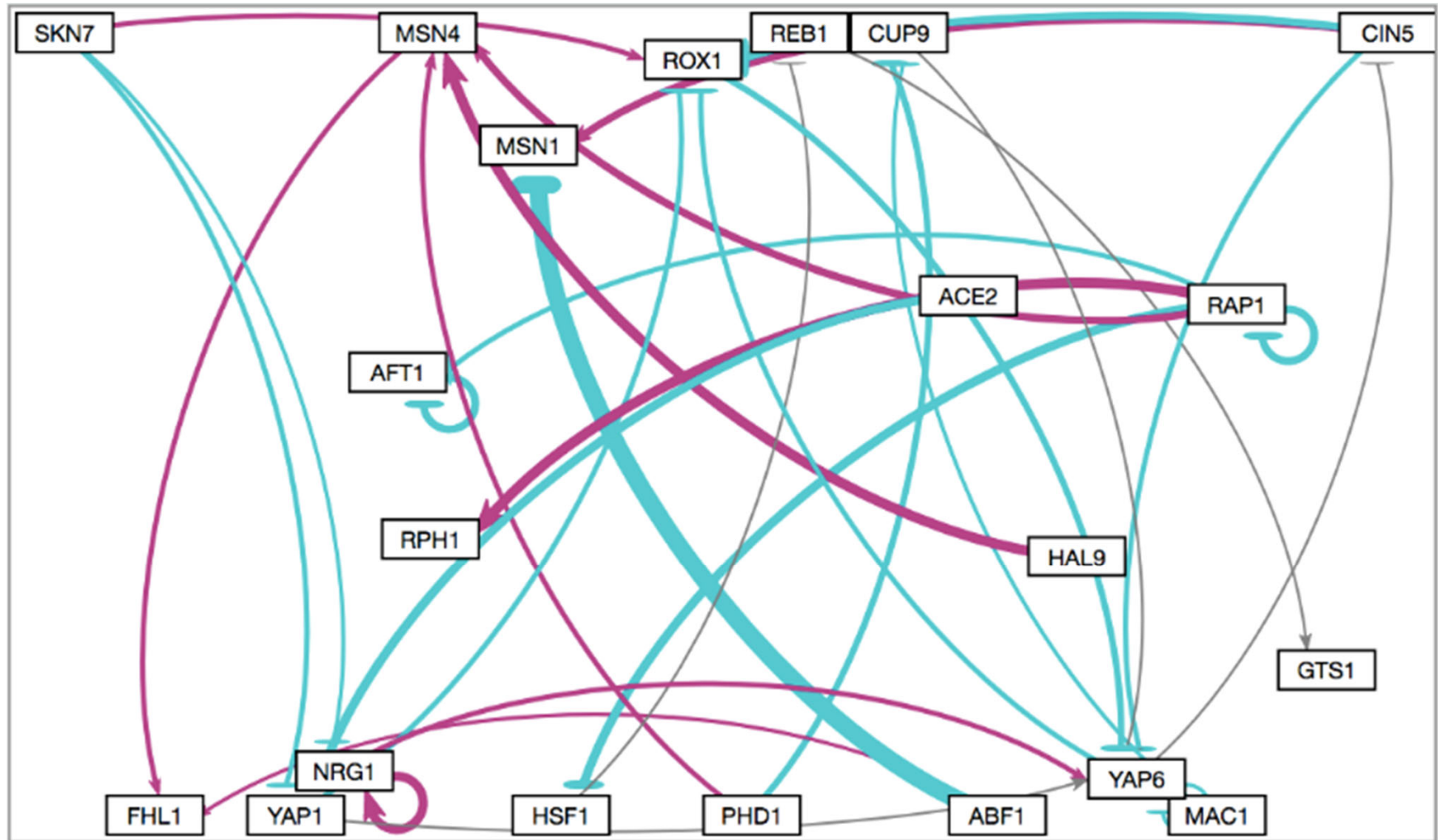
GRNSIGHT V2 (MAY 2017)



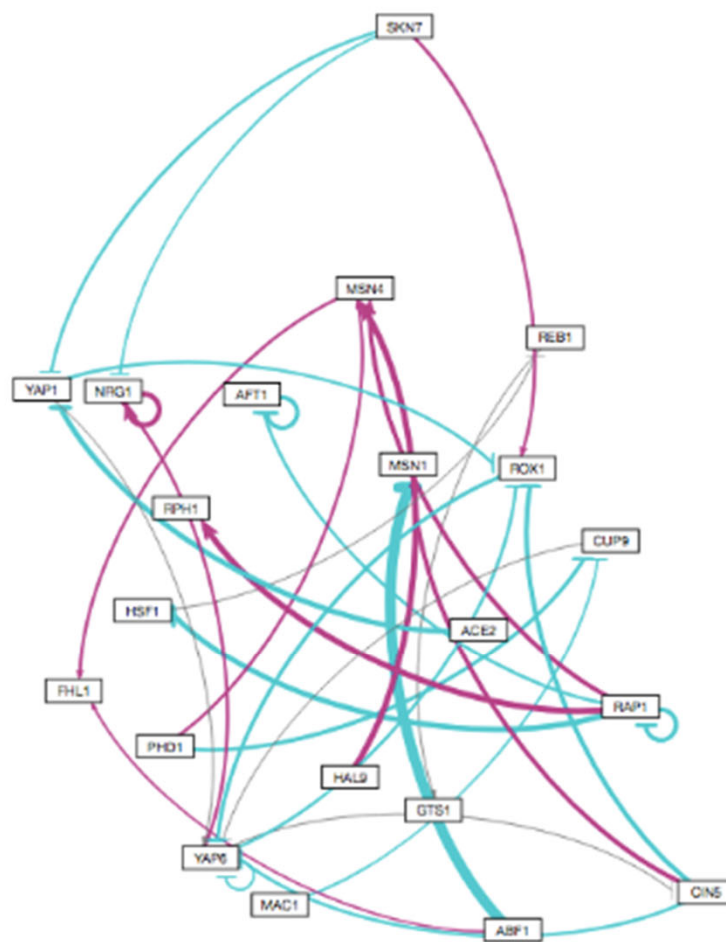
GRNSIGHT V2 (MAY 2017)



GRAPH WITHOUT ADAPTIVE BOUNDING BOX



GRAPH WITH ADAPTIVE BOUNDING BOX



EVERY PROJECT COMES WITH ITS CHALLENGES, AND GRNSIGHT WAS NO DIFFERENT

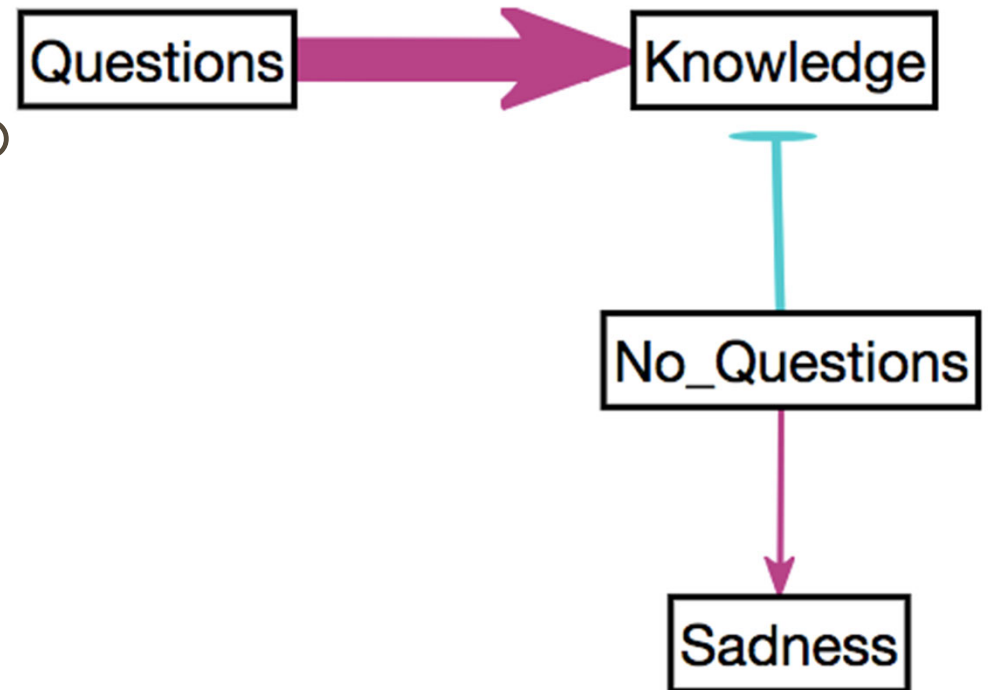
- Working in a team (both a challenge and a blessing)
- Messy, un-commented chunks of code
- Dynamic resizing algorithm
- Windows OS
- Chrome/Firefox updates
- Sudden bug where the entire program broke entirely for two of the developers appeared a week before the end of the semester

SUMMARY

- GRNsight is a web app and service for visualizing small- to medium-scale gene regulatory networks (GRNs).
- New features for v2 include:
 - Separation of viewport from graph bounding box
 - Choice of three viewport sizes
 - Best viewport size automatically detected from browser
 - Viewport can also be fit to window
 - Choice of fixing graph inside viewport or adaptively resizing graph bounding box based on graph size
 - Zooming and scrolling enabled

ACKNOWLEDGMENTS

- Dr. Kam D. Dahlquist
- Dr. John David N. Dionisio
- Britain Southwick
- GRNsight Team:
 - Anu Varshneya
 - Mihir Samdarshi
 - Jen Shin
 - Eddie Bachoura
 - Eileen Choe



GRNsight: <https://dondi.github.io/GRNsight/>

Github: <https://github.com/dondi/GRNsight>

PeerJ Paper: <https://peerj.com/articles/cs-85/>